HTML is not what you think it is.

What is HTML?

HTML stands for Hyper Text Markup Language; it is used to structure and describe the content. It is a Markup language, not a programming language because it does not perform logical operations or complex coding tasks. It defines the layout and elements of a webpage.

Where HTML is commonly used.

HTML is not only used for websites. It can be used anywhere structured content needs to be displayed in a browser or browser-like environment. However, its main and most common use is still the web, while other uses are extensions of the same ecosystem.

Some common uses of HTML are:

1. **Websites:** HTML is mainly used to create and structure websites. It defines elements such as headings, paragraphs, images, links, tables, forms, video, audio, and buttons that appear on a webpage.
2. **Web Applications:** HTML is also used in web applications like online editors, social media platforms, banking systems, and dashboards.
3. **Email Templates:** HTML helps create formatted emails with colors, images, buttons, links, and responsive layouts for marketing and communication purposes.
4. **Documentation & E-books:** HTML is used to create digital documents and e-books. Many online manuals, tutorials, and EPUB files are based on HTML.
5. **Mobile & Desktop Apps:** HTML along with CSS and JavaScript is used to build user interface for some mobile and desktop applications. Technologies like Electron and hybrid app frameworks allow developers to create apps using web technologies.

What HTML Can Do Beyond Structuring User Interfaces.

HTML is commonly known for creating the structure of web pages and user interfaces, but HTML can do many more things than simple layout design. Modern HTML supports many features that improve functionality, accessibility, SEO optimization, communication and multimedia integration across the web.

- **Semantic content organization:** Semantic content is achieved using the HTML semantic tags. It describes the meaning and purpose of the content.
  Key semantic tags introduced in HTML5-
  - → <Header>: Used for introductory content or a set of navigation links at the top of a page or section.
  - → <nav>: Specifically for navigation bar.
  - → <main>: Defines the unique, primary content of the document.

- → <section>: It defines a section in a document.
- → <article>: For self-contained content that could stand alone, like a blog post or news story.
- → <aside>: It defines some content aside from the content it is placed in.
- → <footer>: Contains information about the page or section, such as copyright data or contact links.
- → <figure> and <figcaption>: Used to wrap and image or illustration along with its caption.

Using semantic tags is beneficial for both the developer and the users. It improves readability of code, helps search engines understand content and enhances accessibility.

- **Accessibility Support:** Accessibility means designing web content in a way that allows everyone, including people with disabilities, to access. Understand and use it easily. Accessibility can be achieved using the following practices -
  - → Using alternative text for images.
  - → Using form labels properly.
  - → Using semantic structure.
  - → Descriptive link text.
- **Search Engine Optimization (SEO):** SEO is the practice of improving a website's visibility on search engines, such as Google, Yandex, Microsoft Bing, DuckDuckGo etc., to increase traffic. SEO in HTML involves using specific tags and a semantic website structure to help search engines crawl, index and understand a website's content. By using these practices, a website has a better chance of ranking higher in search results. SEO practices-
  - → Using <title> tag to describe the webpage content clearly.
  - → Add meta description to provide a summary of the webpage.
  - → Use semantic HTML elements to improve content structure and SEO.
  - → Maintain a proper heading hierarchy using <h1> to <h6>.
  - → Add alt text to images for better accessibility and search indexing.
  - → Create clean and readable SEO-friendly URLs.
  - → User internal links to connect related webpages within the website.
  - → Add the viewport meta tag to make webpages mobile-friendly.
  - → Use structured data (Schema Markup) to help search engines understand content.
  - → Improve accessibility using proper labels and meaningful HTML structure.
  - → Use canonical tags to avoid duplicate content issues.
  - → Specify the webpage language using the lang attribute in HTML.